

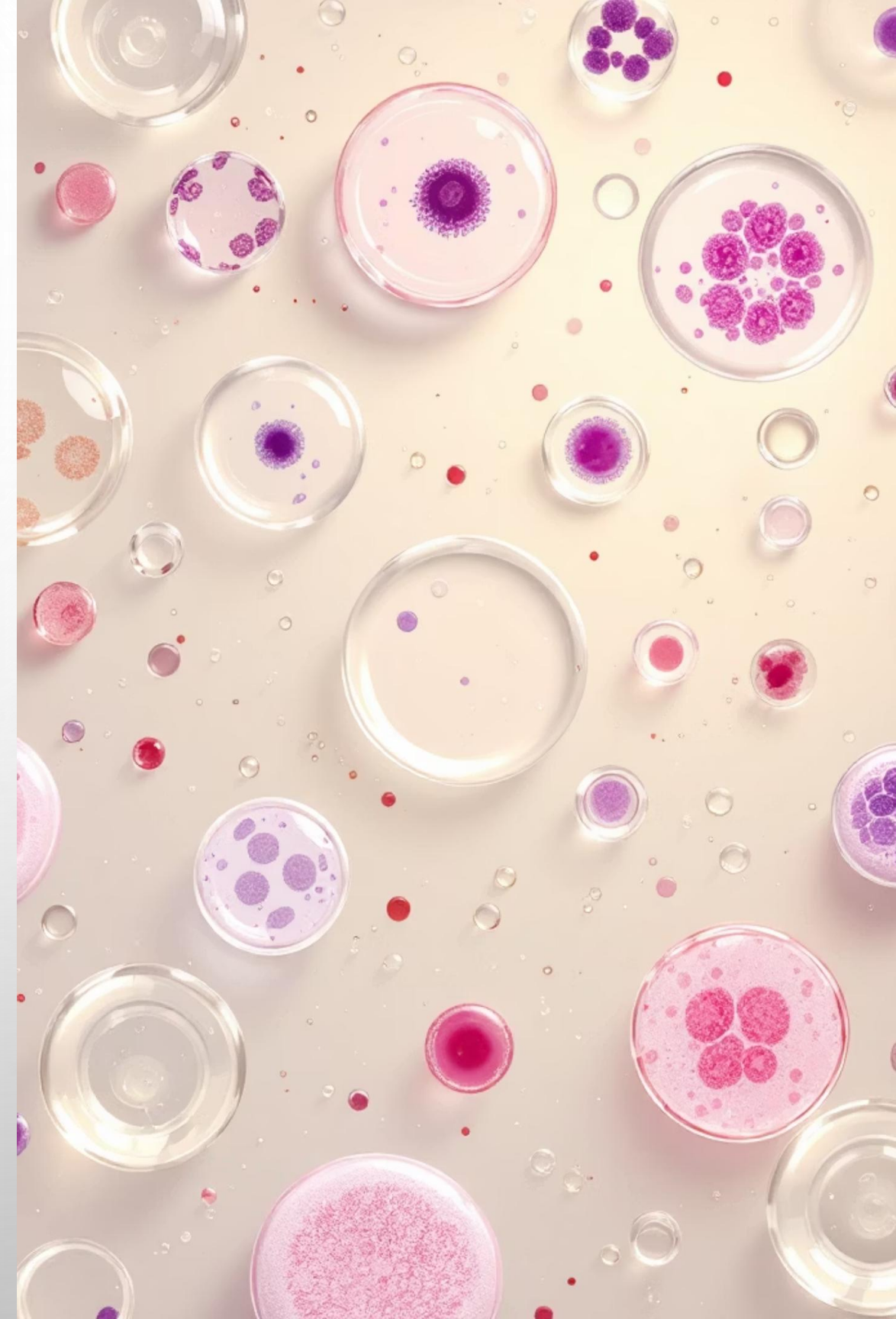
WBC Classification using Deep Learning

Automated White Blood Cell Classification

Mature

Immature

Rare



Problem + Goal

Problem

- Manual WBC classification is slow and error-prone
- Requires expert analysis

Goal

- Build an AI model to classify WBC images automatically

Output Classes

Mature

Immature

Rare

Dataset Overview

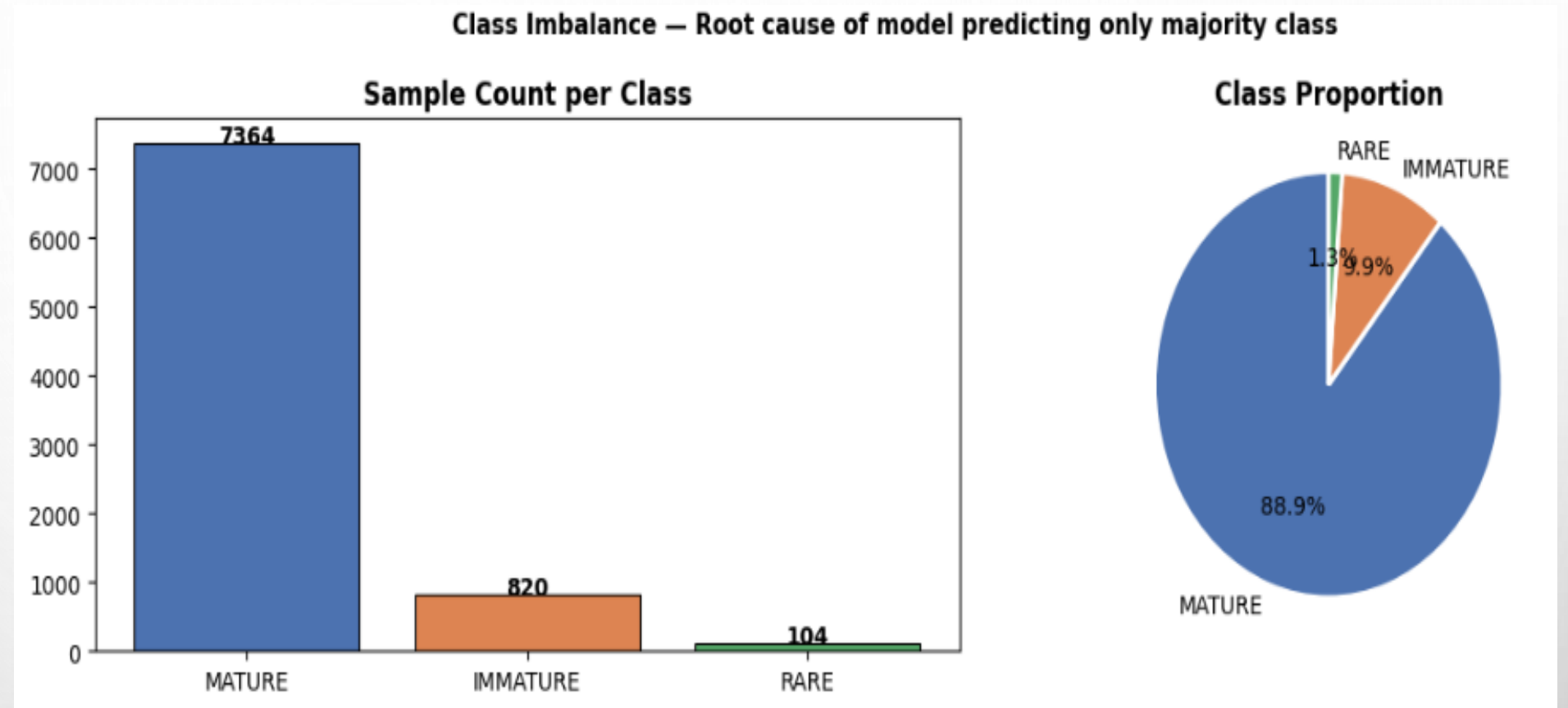
- Total images: ~8,288
- Original labels: 13 → Converted to 3 classes

Class Distribution

- Mature → ~70%
- Immature → Moderate
- Rare → ~5%

Key Issue

- Severe class imbalance

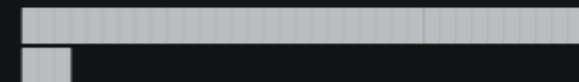


```
0 → MATURE    ( 7364 samples)
1 → IMMATURE   (  820 samples)
2 → RARE       (  104 samples)
```

Total: 8288 samples

Class distribution:

```
MATURE    enc=0    7364 ( 88.9%)
IMMATURE   enc=1     820 (  9.9%)
RARE       enc=2     104 (  1.3%)
```



Exploratory Data Analysis

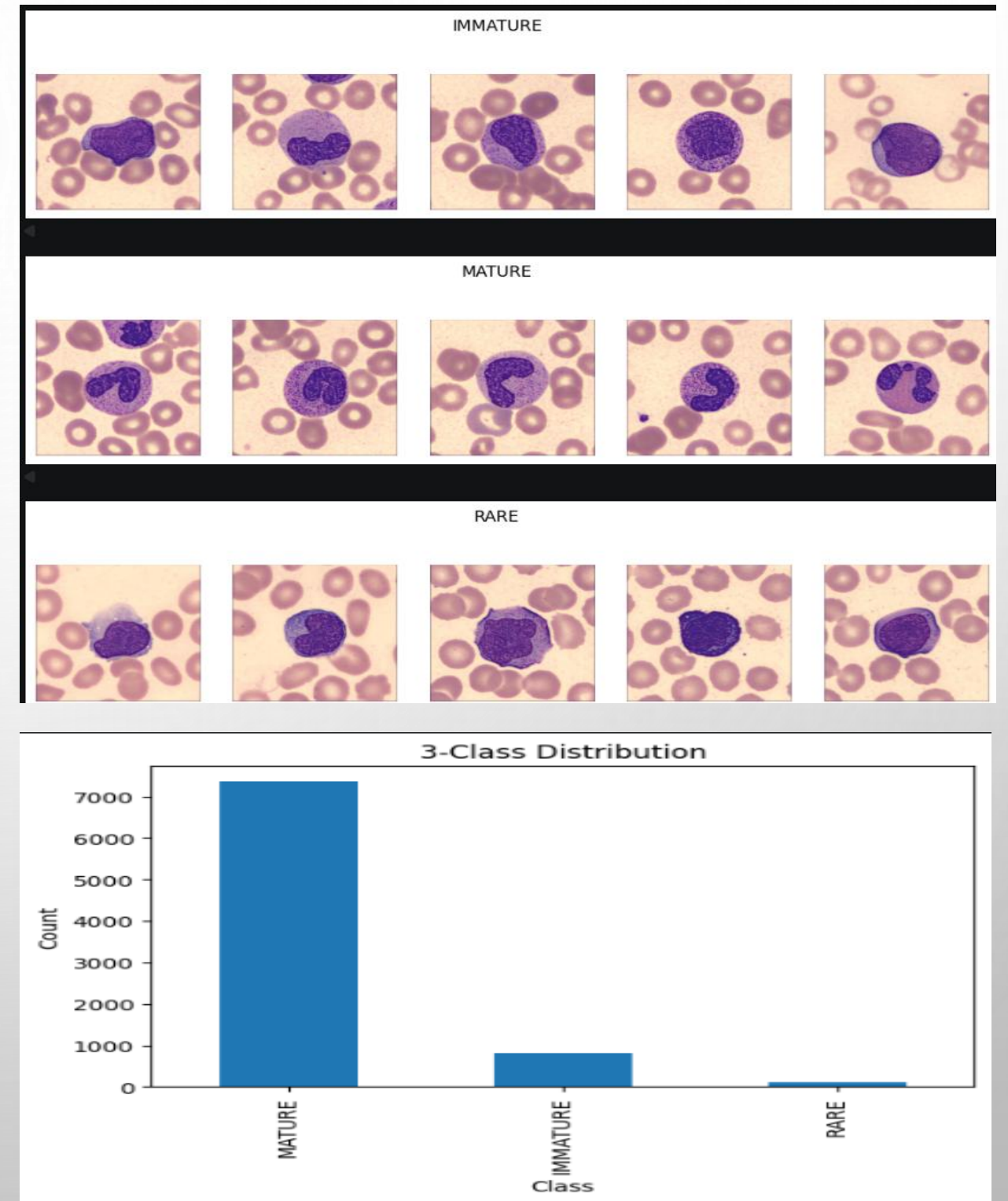
EDA Steps

- Visualized sample images
- Checked class imbalance
- Identified noise

Observations

Background Noise
RBCs, stains present in images

Rare Class
Very limited samples available



Baseline: CNN Failure

Initial Model: CNN

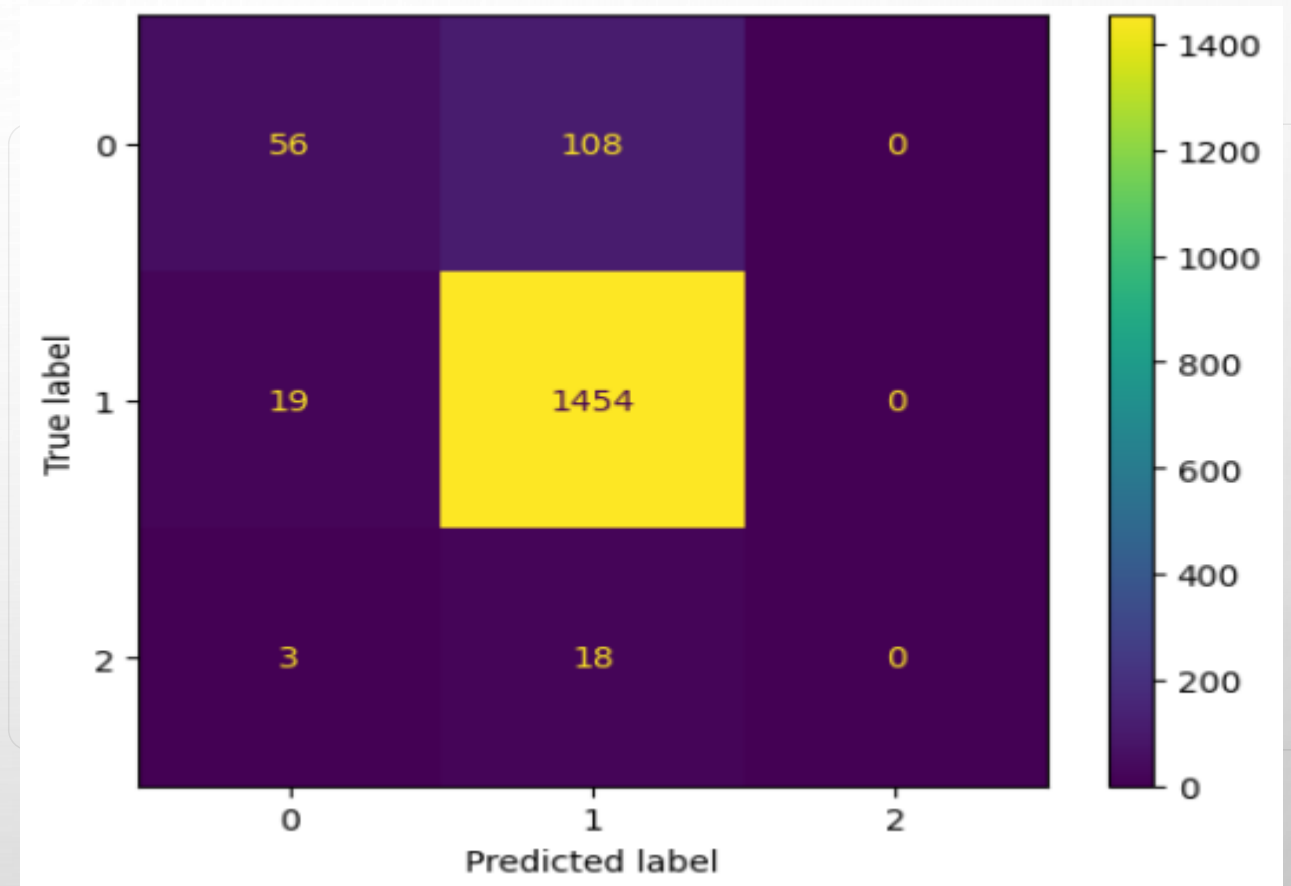
Basic Conv → Pool → Dense

Result

- Predicted mostly **MATURE**
- Rare class = **0** predictions

Reason

Model learned majority shortcut



CNN Confusion Matrix

MODEL 2: EFFICIENTNET-B1

EfficientNet-B1

- Pretrained model
- Scales efficiently

Result

- Better than CNN
- Still weak on rare class

MODEL 3: DENSENET121

DenseNet121

- Dense connections
- Feature reuse

Why it worked

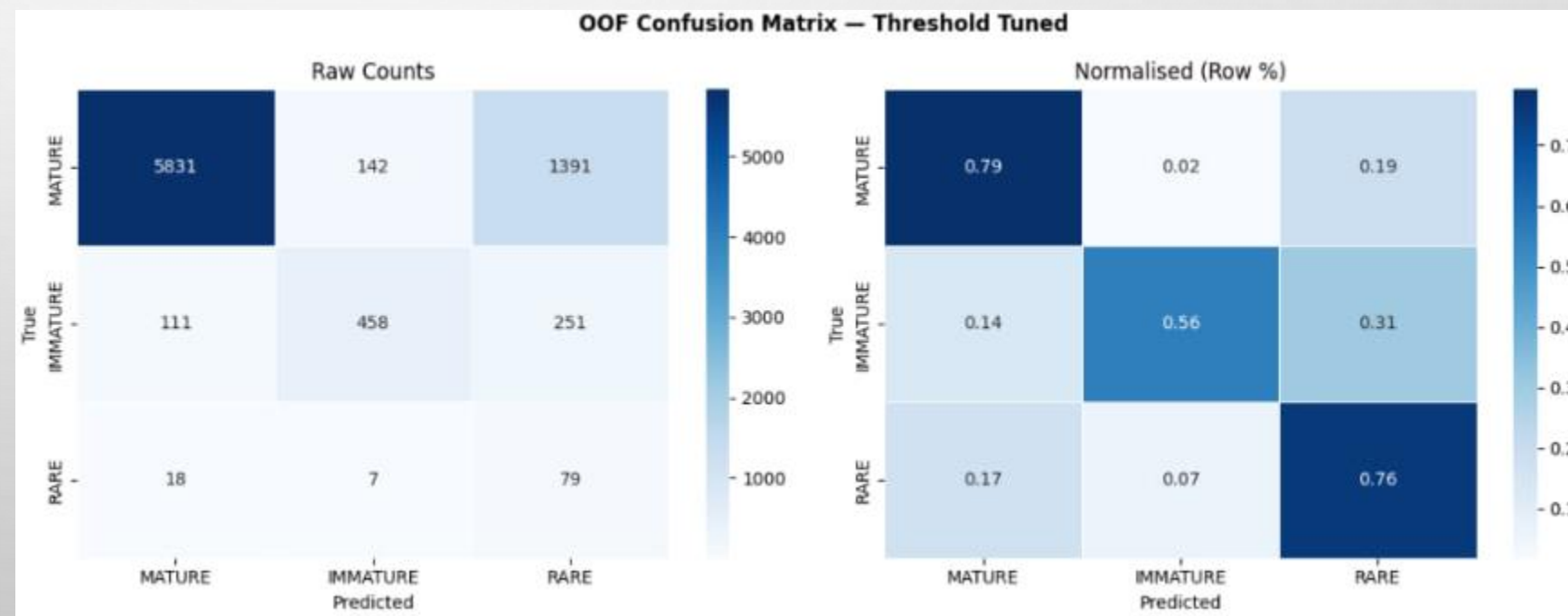
- Preserves fine details
- Good for small datasets

This is the **OOF (Out-of-Fold) combined confusion matrix** — the most honest overall result on 100% of data with zero data leakage. Shows RARE = 0.76 detection. This is your final result slide image.

Efficientnet-B1 Confusion matrix



DenseNet121 Confusion matrix



Major Issues Identified



Class Imbalance

Skewed distribution across Mature, Immature, and Rare classes



Rare Class Ignored

Model failed to learn rare cell patterns entirely



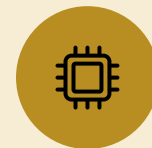
Noisy Images

Background artifacts like RBCs and stains interfered with learning



Slow Preprocessing

Processing pipeline was a bottleneck during training



Inefficient Training

GPU underutilized, leading to long training times

Approach: Full Solution



ROI Cropping

HSV masking to isolate WBC region



Data Augmentation

Expand minority class samples



Class Weights

Penalize majority class dominance



Focal Loss

Focus on hard-to-classify samples



Transfer Learning

Leverage pretrained model features



Preprocessing Optimization

Cache and pre-crop for faster training

ROI Cropping

Purple Masking (HSV)

Extract only WBC region

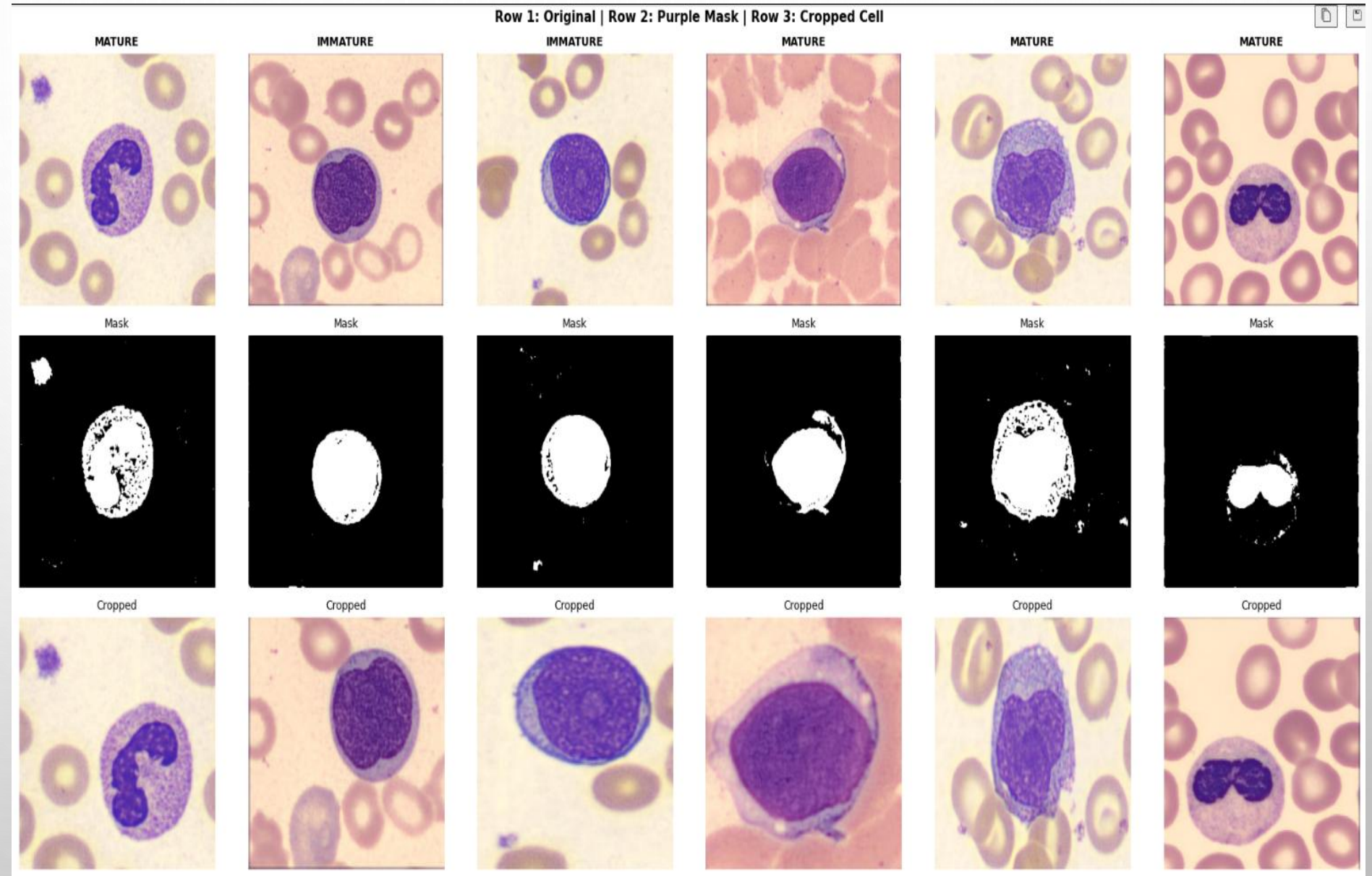
Remove background noise

Why HSV

Stable color detection

Impact

Better feature focus



Data Augmentation

Purpose

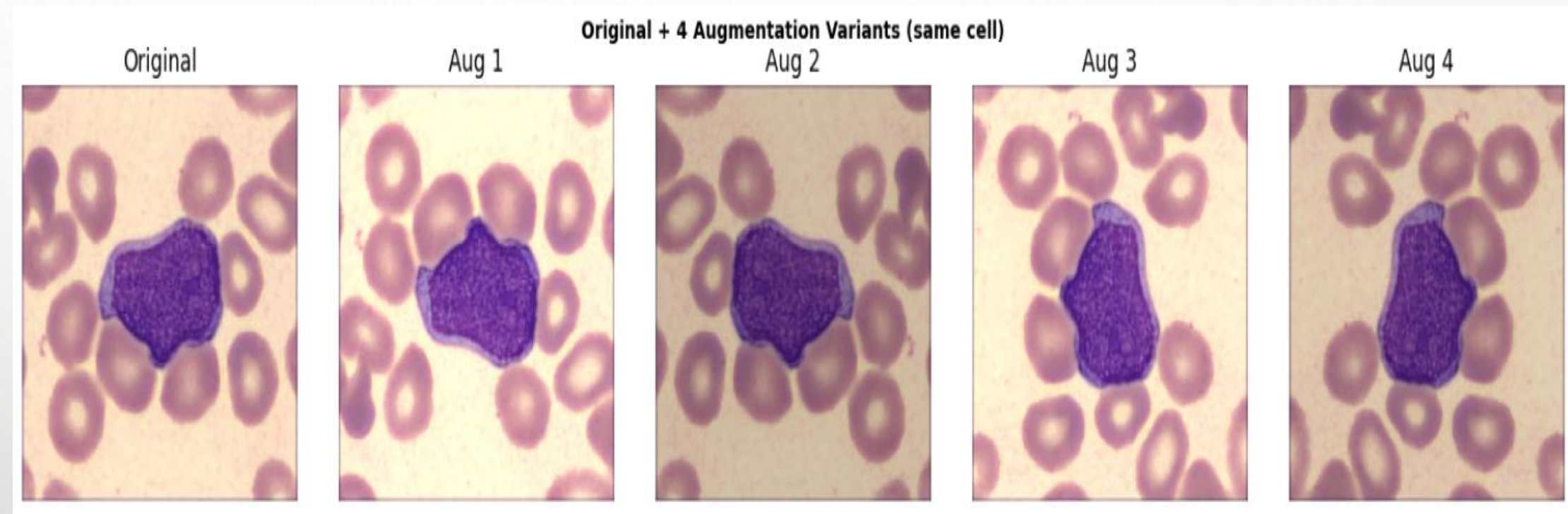
- Increase dataset diversity
- Reduce overfitting
- Handle class imbalance

Techniques

- Flip (Horizontal, Vertical)
- Rotation (0° – 270°)
- Brightness adjustment

Impact

- Better generalization
- Improved robustness
- Helps rare class learning



Original vs Augmented Samples (Same Cell)

Handling Class Imbalance: Focal Loss & Class Weights

Problem

- Model biased toward majority class (mature)
- Rare class ignored during training

Class Weights

- Assign higher importance to Rare class
- Penalizes misclassification of minority class

Effect

- Forces model to consider all classes equally

Focal Loss

- Reduces impact of easy samples
- Focuses on hard / misclassified samples

Key Idea

- Easy predictions → low contribution
- Hard predictions → high contribution

Impact

- Improves rare class learning
- Reduces majority class dominance
- Better balanced performance

Handling Class Imbalance

1

Data Augmentation

- Flip, rotate, zoom

2

Class Weights

- Rare class given higher importance

3

Focal Loss

- Focus on hard samples

Result: Improved rare class learning

MODEL COMPARISON

Model	CNN	EfficientNetB1	DenseNet121
Type	Basic	Pretrained	Pretrained
Feature learning	Weak	Good	Strong
Rare class	Failed	Weak	Better
Small dataset	Poor	Moderate	Best
Verdict	Not suitable	Moderate	Best

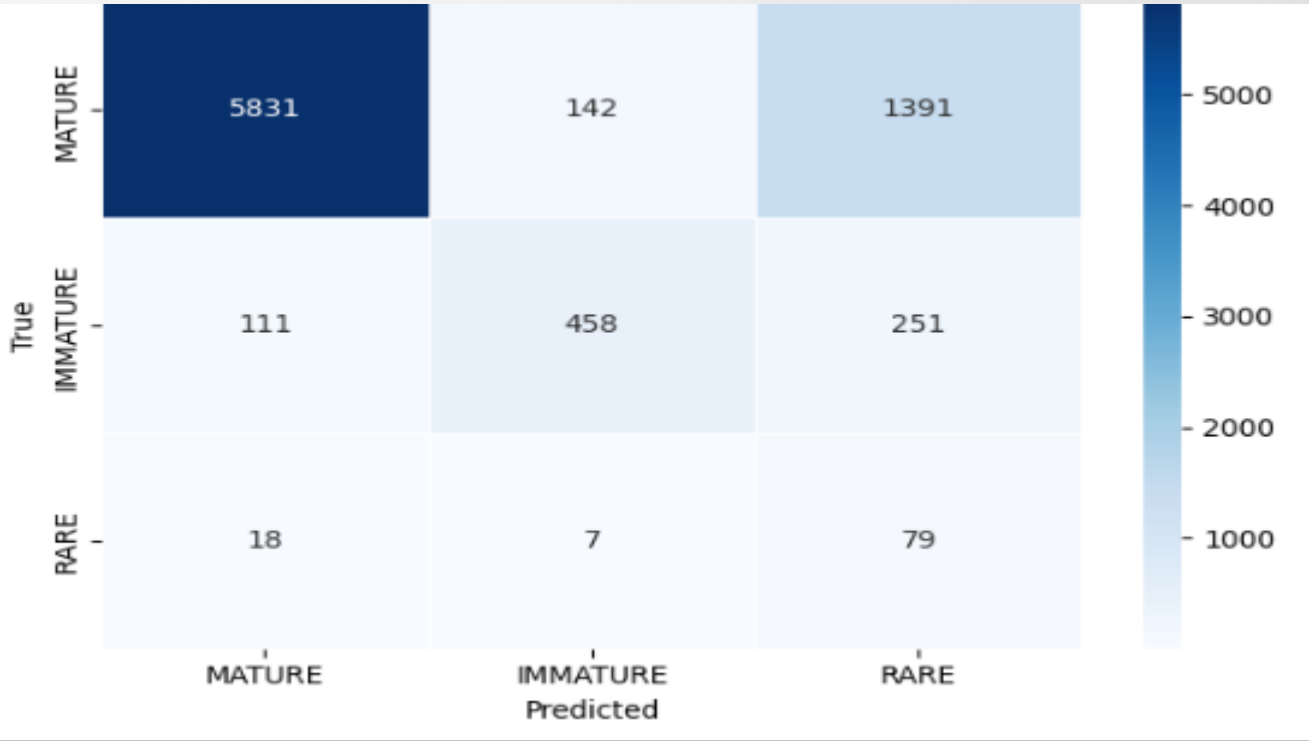
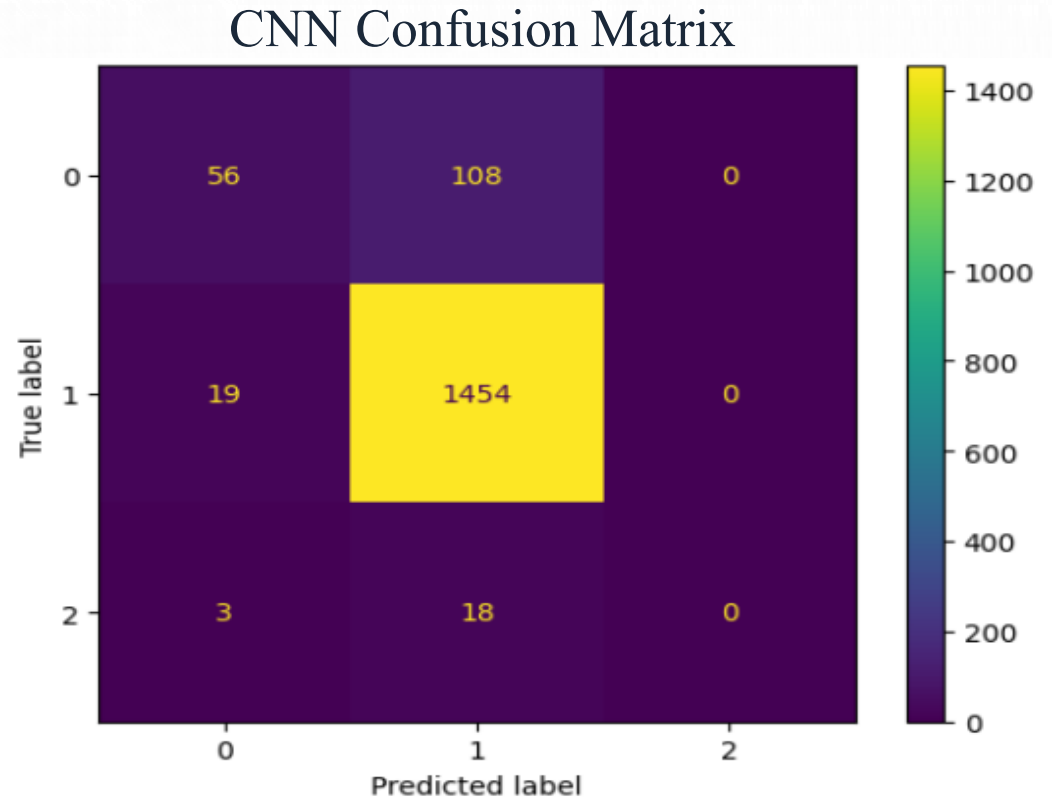
RESULTS

Observations

- CNN → biased predictions
- EfficientNet → slight improvement
- DenseNet → best performance



Efficientnet-B1 Confusion matrix



DenseNet121 Confusion matrix

METRICS

Evaluation Metrics

- Accuracy → misleading
- Balanced Accuracy → correct metric

Why

- Treats all classes equally

Training Curves

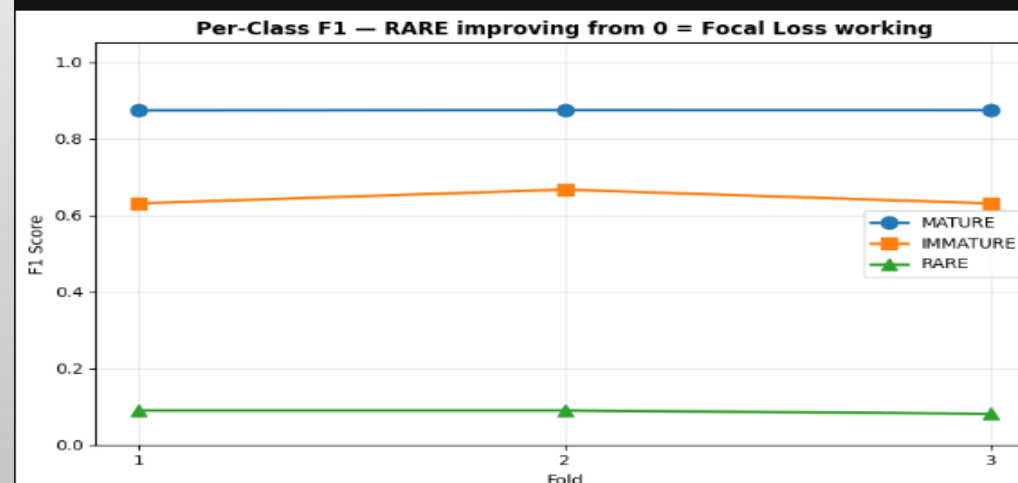
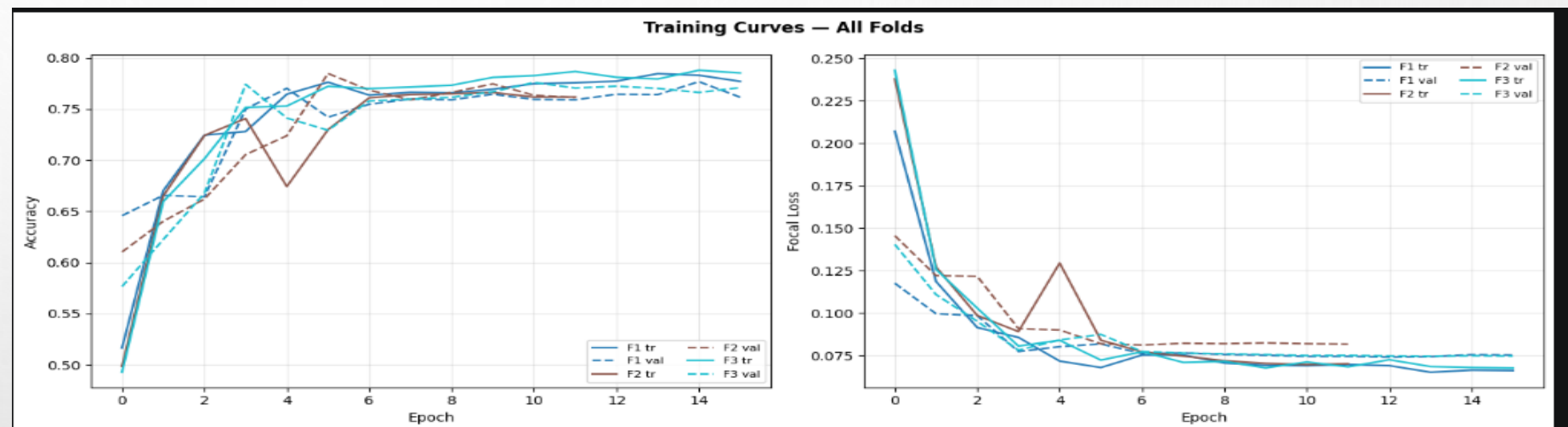
Training accuracy/loss curves across all 3 folds — shows the model learning over epochs and not overfitting badly.

OOF Summary (training data, zero leakage):

Accuracy : 0.7676
Balanced Accuracy: 0.7085 ← key metric
F1 Macro : 0.5346

Per-Class F1:

MATURE : 0.8742
IMMATURE : 0.6422
RARE : 0.0872



CONCLUSION

Final Outcome

- DenseNet121 performed best
- Class imbalance is main issue
- Rare class still challenging

Future Work

- More rare data
- Better augmentation